HBA PROGRAM: **Decision Making with Analytics (DMA)**

BUSINESS 3316

FACULTY TEAM

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INTRODUCTION

Business 3316 teaches you how to use data and models to make better managerial decisions. Working with real business cases, you'll learn core quantitative tools (decision trees, probability, linear regression, simulation, and optimization), when to use them, and how to combine them.

Analytics however is more than a set of tools. It is a way to frame problems, test assumptions, and quantify uncertainty. You'll practice choosing the right method for each decision and communicating what the numbers say from a manager's point of view.

We apply these ideas across marketing, operations, finance, accounting, and strategy. Examples include evaluating channels and new products, sizing plants and inventory, optimizing portfolios and transportation routes, and setting effective prices.

The major segments of the course are:

1. Decision Making under Uncertainty and Analyzing Sequential Decisions
2. Building Spreadsheet Models
3. Data-driven Decision Making and Linear Regression
4. Simulating the Future
5. Simultaneous Decision Problems and Optimization

LEARNING OUTCOMES

At the end of the **Decision Making under Uncertainty and Analyzing Sequential Decisions** module, students will be able to:

- Understand probability, possible outcomes, decision points and uncertain events
- Illustrate decision problems using decision trees with sequences of decisions and/or uncertain events
- Calculate expected values and risk profiles for decision alternatives in decision trees (with or without embedded decisions)

At the end of the **Building Spreadsheet Models** module, students will be able to:

- Construct conceptual and/or spreadsheet models of business decision problems
- Make reasonable, justifiable assumptions to abstract complex problems
- Manipulate inputs and conduct sensitivity analysis to evaluate impacts on the decision

At the end of the **Data-driven Decision Making and Linear Regression** module, students will be able to:

- Calculate and explain the average and standard deviation of historical data
- Build and interpret a histogram
- Summarize data using averages, standard deviations, and histograms
- Perform and critically evaluate single- and multi-variable regression (with continuous, categorical, and interaction variables), including interpretation of coefficients, uncertainty, and diagnostics
- Use regression models to analyze relationships among variables and to develop predictions

At the end of the **Simulating the Future** module, students will be able to:

- Identify and model sources of uncertainty in business problems using probability distributions (e.g., normal, uniform, empirical)
- Construct and validate simulation models
- Use simulations to estimate expected values, generate risk profiles, and conduct scenario analyses

At the end of the **Simultaneous Decision Problems and Optimization** module, students will be able to:

- Formulate optimization problems as linear or nonlinear programs by specifying decision variables, objective function, and constraints, and recognize when a problem is nonlinear
- Solve optimization models using Excel Solver (including linear, integer, and binary constraints)
- Interpret solution outputs, including shadow prices and sensitivity ranges
- Assess the robustness of solutions to changes in constraints or objective function coefficients
- Represent and analyze multi-objective decision problems using efficient frontiers

Throughout **each module**, students will be able to:

- Articulate analyses and conclusions clearly in class discussions, written assignments, reports, and presentations
- Develop and communicate well-reasoned recommendations that connect analytical findings to decision making

MATERIALS

Cases and Textbook

1) Casebook for Business 3316
2) Bell and Zaric (2013). *Analytics for Managers*. Routledge.

Analytics for Managers is available for free through Western Library and is linked on our course website.

Software

This course requires Microsoft Excel for Windows (Microsoft 365 desktop app) with the Data Analysis Tool Pack and Solver add-ins.

Ivey students may download Microsoft Office free of charge at myoffice.uwo.ca.

Important for Mac users:
Excel add-ins on Mac are unreliable and may produce errors. To avoid problems, use Excel for Windows via Boot Camp (recommended) or virtual software (e.g., Parallels). More details: Boot Camp Installation Guide

Using Excel on Mac is at your own risk: past issues have negatively affected grades, and faculty cannot provide support for versions other than the required Windows version.

OVERALL GRADING SCHEME

The grade in this course is a composite of class contribution, a midterm exam and a final exam. I will assign your grades for written and non-written work based on the following weights:

Course Component	Weight
Contribution	25%
Midterm Exam	35%
Final Exam	40%
Total	100%

DMA CLASS CONTRIBUTION GRADING SCHEME

We briefly describe our course-specific contribution policies. HBA program policies (which also apply) are covered below.

Your class contribution grade will be based on a session-by-session assessment of your performance, with the lowest two non-negative scoring sessions ignored. The Attendance and Professionalism component may also reflect in-class or pre-class submissions required.

Component	Weight	Notes
Attendance and Professionalism	50%	1 point for each session that you were on time, prepared, and not disruptive. (Max of 23 points)
Quantity	20%	1 point for each session in which you have spoken. (Max of 10 points)
Quality	30%	1 point = above average contribution to class 2 points = outstanding contribution to class This component has no maximum and will be scaled to achieve an average class contribution grade of 80.

OFFICE HOURS/APPOINTMENTS

Each faculty member will provide guidance on their weekly office hours; communicated in class, via Learn or via email.

HBA PROGRAM POLICIES

EXAMS AND MAKEUP EXAM POLICY

At Ivey, exams are more than assessments—they are reflections of professional readiness and academic growth. This policy upholds our values of Integrity, Courage, Inclusivity, and Community, and outlines a fair, transparent process for accessing makeup exams when unexpected challenges arise.

Policy Statement

Examinations are a core component of academic evaluation and professional development. Students are expected to treat exam schedules with the same seriousness as professional obligations.

In exceptional circumstances, students may request academic consideration in accordance with the **Academic Consideration Policy**.

Academic Consideration for Missed Exams

- Students must **formally request academic consideration** as soon as possible—ideally before the scheduled exam.
- Requests must be **supported by appropriate documentation** and submitted within **48 hours** of the missed exam.
- Students should communicate **proactively** with their Academic Advisor to clarify requirements and access support.

Makeup Exam Policy

- Students **granted academic consideration** will be offered **one official makeup exam opportunity**.
- The makeup exam will be scheduled within the **same academic term**, where possible.
- Students must prepare fully and approach the makeup with professional responsibility.
- **Missing the makeup exam** will result in **unsatisfactory academic standing**.
- A **second makeup exam** will be authorized **only at the Dean's discretion** after a formal review.
- Unsatisfactory standing will **remain in effect** until the review is complete and lifted only if the second makeup is granted and passed.

CLASS ATTENDANCE AND CONTRIBUTION POLICY

This policy reflects Ivey's shared values of Integrity, Courage, Inclusivity, and Community. Attendance is more than a rule – it is a commitment to collaborative learning, shared accountability, and personal growth. In a case-based environment, presence and engagement are essential to both individual success and community learning.

Students are expected to attend all scheduled classes and participate meaningfully, demonstrated through respect for classmates and instructors. Consistent, respectful engagement is a critical part of professional conduct and directly supports learning and progression. See the Contribution Grading Expectations section for examples of contribution.

Attendance Expectations

- Attendance is mandatory in most courses and detailed on the syllabus

- Students who miss more than five (5) 80-minute sessions may:
  - be placed in unsatisfactory standing
  - be ineligible to write final exams or assessments, regardless of academic consideration
- All absences – including those granted academic consideration – count toward the total, **except for absences due to religious observance**.
- Students nearing the 5-class absence threshold will receive email notification from their faculty, with their Academic Advisor copied. Students are encouraged to seek support.
- Students who reach 50% of the threshold before or by reading break will be enrolled in a learning strategies course at the discretion of the Faculty Director and Director of Culture, Inclusion, Student Support and Engagement.

Incomplete Attendance (e.g. lates)

- '**Late is Late'** – students arriving after discussion has begun, or who depart class before discussion has finished will be counted 'late.' Students will receive a contribution deduction (-1 point or 0 for the day)
- At the discretion of the Instructor, students who leave and return to class may not be invited into later discussion. These students have missed important sections of discussion and their comments may replay points made while they were out of class. To the extent possible, please use time between classes for breaks, calls, and other activities.
- Departing class and returning with food or beverages may be counted as disruptive/disrespectful behaviours.
- Students who miss more than 20 minutes of any class period should be counted as being absent.

Academic Consideration for Attendance

At Ivey, we recognize that personal and academic challenges may arise unexpectedly. This policy provides a fair, timely, and compassionate process for academic consideration—rooted in our values of Integrity, Courage, Inclusivity, and Community. It supports student well-being while maintaining academic standards.

This policy applies when a student:

- Misses a **non-contribution-based graded component** (e.g., exam, assignment, in-class test, group work) due to unforeseen illness or life events.
- Misses a component due to **Varsity or Case competition participation**, or **religious observance**.
- Experiences an **extended absence** impacting academic engagement.

Valid Grounds for Consideration

- **Catastrophic Events** – Bereavement, major accidents, etc.
- **Illness (Physical or Mental Health)** – Including chronic, acute, or temporary conditions.
- **Religious Observance** – Conflicting with academic obligations.
- **Varsity Commitments or Ivey Case Competitions** that conflict with assessments.

Important Notes

- All absences—including those with academic consideration—count toward total missed classes, **except religious observance**.
- Students nearing the 5-class threshold will be notified by faculty and copied to their Academic Advisor.

- Students who reach 50% of the threshold by reading break will be added to a learning strategies course at the discretion of the Faculty Director and Director of Culture, Inclusion, Student Support and Engagement.
- **Excessive absences may lead to unsatisfactory academic standing**, even with academic consideration.

Submission Process

Step	Action	Who	Deadline
1	Report absence (Varsity/Religious) via LEARN & submit form/documentation	Student	≥10 days before event Students must submit supporting documentation with this request 10 days prior to the planned varsity event or Religious Observance.
2	Report absence (Illness/Catastrophe) via LEARN + Western Medical Documentation Form	Student	Within 48* hours of absence *A Western Medical Certificate needs to be obtained as soon as the student becomes ill, substantiating why the student was medically unfit to attend class on the assessment date.
3	Review & Verification	Academic Advisor	Within 3 business days
4	Receive Decision	Student	Upon completion of review

Academic challenges are part of learning. This policy supports fairness while encouraging resilience, self-advocacy, and collaboration between students, faculty, and advisors.

Student Policy Responsibility Table

Scenario	Action
Illness or emergency?	Submit a request in LEARN within 48 hours
Varsity conflict or religious event?	Submit documentation in LEARN 10+ days in advance
Have questions?	Contact an Academic Advisor
Disagree with a decision?	Speak to an Academic Advisor for further steps

CONTRIBUTION GRADING EXPECTATIONS

The following list might be of help regarding expectations in terms of contribution:

Outstanding Contribution (90-100%)

- Contributes at the highest levels throughout the class;
- Contributions indicate thorough preparation;
- Contributions reflect careful attention to the ongoing discussion;
- Contributions include strong follow-up comments and questions;
- Helps integrate course concepts with other subjects;
- Frequently explains difficult points or concepts;
- Provides sustained strong support to enrich classroom learning;
- Builds and develops outstanding arguments for a position;
- Arguments include strong, valid, and relevant evidence.

Excellent Contribution (80-89%)

- Contributes frequently to discussions;
- Contributions indicate careful preparation;
- Contributions reflect attention to and consideration of the ongoing discussion;
- Explains difficult points and concepts;
- Helps guide class discussions;
- Regularly builds and develops strong arguments for a position;
- Arguments include strong evidence.

Good Contribution (70-79%)

- Contributes fairly regularly to discussions, but primarily in the beginning of the class;
- Contributions indicate consistent preparation;
- Contributions reflect thoughtful responses to the ongoing discussions;
- Raises questions that help focus on difficult points or concepts;
- Occasionally builds reasonable arguments for a position;
- Arguments sometimes include valid evidence.

Fair Contribution (60-69%)

- Contributes irregularly to class discussions;
- Gives occasional indication of preparation;
- Rarely builds on the ongoing discussion;
- Positions rarely supported by valid evidence.

Poor Contribution (50-59%)

- Contributes rarely to discussions;
- Gives little indication of preparation and thought;
- Contributions confuse statements of opinion with statements of fact;
- Does not aid in providing a positive atmosphere for meaningful discussion.

Unsatisfactory Contribution (0-49%)

- Actively inhibits or impedes the course of class discussions;
- Misses a number of class sessions;
- Disruptive or unprofessional behaviour.

CLASSROOM NORMS & VALUES

In our Ivey classrooms, our values guide our decision-making and support our continuous pursuit of excellence:

INTEGRITY
We expect honesty and accountability in our actions and decisions, seeking to inspire ethical and responsible leadership.

INCLUSIVITY
We respect diversity of thought, identity, and perspective, recognizing that inclusive communities foster innovation and creativity.

COMMUNITY
We see strength in positive and supportive interactions, take pride in our collective accomplishments, and have confidence in our long-term success.

COURAGE
We pursue ambitious goals through our commitment to academic and research excellence and believe that, through an entrepreneurial spirit and capacity to act, we can achieve meaningful outcomes locally and globally.

Practicing these values will help foster a community where members contribute to a supportive, inclusive, and accountable culture that pursues excellence, is aligned with our values, and is nurtured through curiosity, respect, mentorship, and recognition.

EXPECTATIONS FOR PARTICIPATING IN OUR LEARNING ENVIRONMENT

Attendance: Please review Ivey's Class Attendance and Contribution policy above.

Punctuality: Late arrival to class is distracting for faculty and students and detracts from our learning environment. Class times are posted well in advance and available through your course tiles on Learn. Please plan ahead and know when your class is in session. To the extent possible, use time outside of class for other activities (breaks, washrooms, calls, etc.).

Technology in the classroom: In the first session of each course, your professor will establish guidelines and expectations for technology use in their course. Given the differences in course content, delivery and application, technology use will vary between courses. You may observe differences in how your classmates engage with, or access technology in the classroom, from yourself. This is because we understand that to have open and accessible classroom spaces, we may need to proactively accommodate students diverse learning needs through access to technology. Accommodations for learning differences support equity in our learning environments, so that students can engage, make sense of and contribute in our classroom spaces. For students who require access to learning accommodations related to technology use, please contact Western's Accessible Education team (http://academicsupport.uwo.ca/accessible_education/index.html).

Respect for our academic community: Please note that video or audio recording of classes, guest presentations, events, and other academic activities is strictly prohibited without express written permission of your faculty or Program Services office (in instances outside of your course).

ACADEMIC OFFENSES

Scholastic offenses are taken seriously and students are expected to read the appropriate policies at:

https://www.uwo.ca/univsec/pdf/academic_policies/appeals/scholastic_discipline_undergrad.pdf

CODE OF CONDUCT POLICIES

- Western Code of Student Conduct: http://www.uwo.ca/univsec/pdf/board/code.pdf
- Ivey Student Code of Professional Conduct located in the Student Handbook.
- Western Non-Discrimination/Harassment Policy: https://www.uwo.ca/univsec/pdf/policies_procedures/section1/mapp135.pdf
- The Code of Behavior for Use of Computer Resources and Corporate Data: https://www.uwo.ca/univsec/pdf/policies_procedures/section1/mapp113.pdf

PLAGIARISM

Students must write their essays and assignments (at Ivey this includes case exams and reports) in their own words. Whenever students take an idea, or a passage from another author, they must acknowledge their debt both by using quotation marks where appropriate and by proper referencing such as footnotes or citations. Plagiarism is a major academic offence.

All required papers (at Ivey this includes case exams and reports) may be subject to submission for textual similarity review to the commercial plagiarism detection software under license to the University for the detection of plagiarism. All papers submitted will be included as source documents in the reference database for the purpose of detecting plagiarism of papers subsequently submitted to the system. Use of the service is subject to the licensing agreement, currently between Western University and Turnitin.com (www.turnitin.com).

PROHIBITED INFORMATION

Unless explicit instructions are received from your instructor to the contrary, your preparation must be limited to the information provided by your instructor(s), plus your own experience and that of your class peers. Class contributions should be based exclusively on your preparation and discussions with members of your learning team or section classmates, and not augmented with information obtained anywhere else. Information related to cases that have been obtained from sources other than your instructor in paper, electronic or verbal form is not permitted to be used for class contribution purposes. Such prohibited information includes, but is not limited to, the following:

- Hand-outs, including follow-on ("B") cases
- Exam or report feedback (related to a particular case) for other students or provided in other sections or years
- Notes, spreadsheets, etc. specific to cases (obtained before the case has been taught) prepared by students in other sections or years
- Company information that is not provided in the case (and that is not derived from your own experience)
- Teaching notes

If it is determined that such prohibited information is used in class discussions, such an incident will be deemed to be a violation of Ivey's Student Code of Professional Conduct.

Notes on Use of AI or Other Technology Assistance

Our goal is to cultivate your judgment and wisdom in decision-making and ability to apply knowledge to practical situations. Using generative AI tools can be beneficial for your learning, providing synthesis of vast amounts of information**, of helping explore new ideas, and analyzing data efficiently. However, over-reliance on AI, substituting it for your own critical thinking and cognitive effort, will hinder your ability to develop analytical skills, interpret and evaluate the quality of analyses, regardless of whether that analysis is human- or AI-generated.**

Our expectations are designed to foster good practices that effectively integrate available tools without being replaced by them, ensuring you develop the essential judgment needed in an increasingly complex world. To be business leaders in the future, we expect that you will use tools available to you in an ethical and thoughtful manner.

To this end, we have the following expectations:

Preparation

- You should read cases and develop answers to assignment questions without the use of AI. This effort will help you develop analytical abilities that are critical to effective decision making. Following individual preparation, you may engage in learning teams discussions to further your understanding and analysis of the material. Your instructor may also recommend AI tools to further augment your preparation.
- You should respect the intellectual property rights of copyright holders and not upload copyrighted material without permission to any site.

In-Class

- In general, use of technology devices in class is not allowed. Some courses may allow the use of technology for specific activities. Using technology when not allowed will negatively affect your contribution grade.
- Using AI tools in a class session is expressly prohibited and an academic integrity violation as part of the class contribution requirement. An instructor may make an explicit exception to this rule.

Written Work

- For submitted written work, be clear to acknowledge AI usage explicitly. For simplicity, we consider AI tools to be an external expert who has been consulted by you. In most citation methods, this citation is done as a personal communication or as a footnote at the start of the work that explains the usage.
- You are welcome to use AI tools for background research. However, the final product should represent your own synthesis of the ideas and must be written in your own words. In a group assignment, the group is collectively responsible for the submission. You are responsible for any plagiarism.

HEALTH AND WELLNESS SERVICES

As part of a successful student experience at Western, we encourage students to prioritize their health and wellness. Western provides several on-campus services to help you achieve optimum health and engage in healthy living while pursuing your degree.

Students seeking help regarding mental health concerns are advised to speak to someone they feel comfortable confiding in, such as a faculty member, the Faculty Program Director, or the HBA Program Services team. Students who are in emotional and/or mental distress should also refer to Mental Health@Western (http://uwo.ca/health/mental_wellbeing) for a complete list of options on how to obtain help.